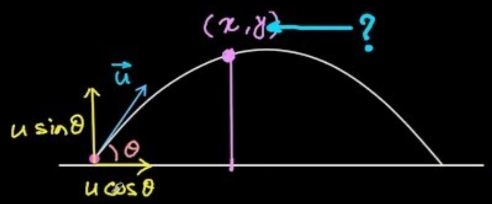
Human-Computer Interaction
Modelling Interaction

Why modelling interaction?

- A model is …
  a simplification of reality

$$y = x \tan\theta + \frac{g x^2}{2u^2 \cos^2\theta}$$



- The architect's model is a **description** of space usage, movement of people, light, shade, etc.
- The physicist's model is a **prediction** of the ball's position as a function of initial velocity and angle.

What can be modelled?

- User intentions, needs and behavior
  - e.g., user goals, task hierarchies, reaction time, error types, learning curve, accessibility needs, etc.
- Cognitive processes
  - e.g., attention & perception, memory & recall, cognitive load, mental models, decision-making processing, etc.
- Physical interaction
  - e.g., pointing & clicking, typing speed & accuracy, touch & gesture, eye movement & gaze, speech input & recognition accuracy, etc.
- Emotional and affective states
  - e.g., frustration or satisfaction, trust in the system, engagement, user motivation
- Context of use
  - e.g., environmental factors, social context, etc.

Models of Interaction

- Descriptive models: to understand
  - Deal with concepts and features.
  - Provide a framework or context for thinking about or describing a problem.
  - Are usually a very simple categorization of features in an interface.

- Predictive models: to measure or forecast
  - Deal with numbers NOT concepts.
  - Are equations.
  - Predict the outcome (accuracy, performance, time) based on a set of predictors.

Descriptive Models

- Any reduction or partitioning of a problem space qualifies as a descriptive model.
- Provide a basis for discussing, understanding, reflecting, and reasoning about certain facts and interactions.
- are used to inspect an idea or a system and make statements about their probable characteristics.
- can reveal flaws in the design and style of interaction

Other names:
Design space, framework, taxonomy, classification

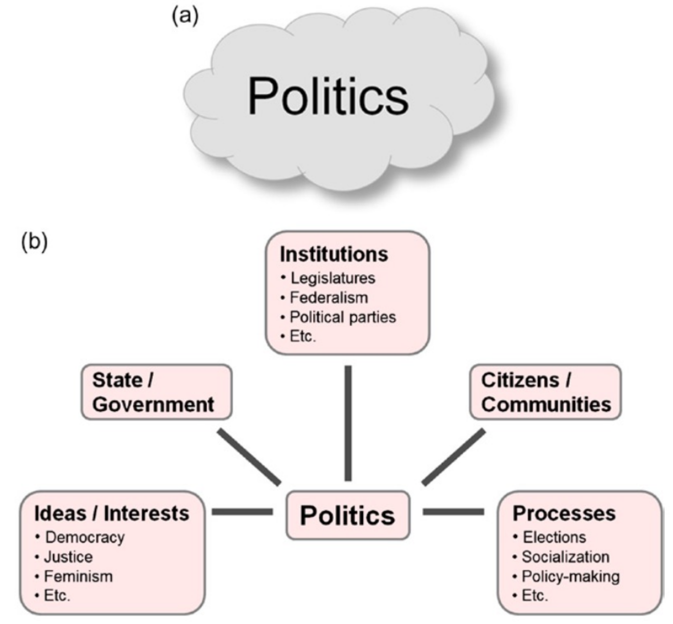


FIGURE 7.1

Politics: (a) The "big fuzzy cloud" model. (b) A descriptive model delineating the problem space.

Examples of descriptive models

- Quadrant model of groupware
  Classifies collaborative systems based on time and place
- Key-action model (KAM)
  Categorizes keyboard keys based on their function during interaction
- Guiard’s model of bimanual control (Two-handed input)
  Describes how humans use both hands asymmetrically during tasks
- Three-state model of graphical input
  Explains pointing device behavior through three states

Quadrant model of groupware (Johansen, 1991)

- Groupware are computer applications that support collaborative work
- Partitions the problem into 2x2 space
  - Location → same place | different places
  - Time → same time | different times

	Same Time	Different Times
Same Place	Copy boards PC Projectors Facilitation Services Group Decision Room Polling Systems	Shared Files Shift Work Kiosks Team Rooms Group Displays
Different Places	Conference Calls Graphics and Audio Screen Sharing Video Teleconferencing Spontaneous Meetings	Group Writing Computer Conferencing Conversational Structuring Forms Management Group Voice Mail

FIGURE 7.2

Quadrant model of groupware.

Key-action model (KAM)

- Keyboards date at least to the 1870s with the introduction of type machines.
- Today's keyboards retain the same letter arrangement (qwerty) but have many other keys too.
- Categorizes keys into three groups.

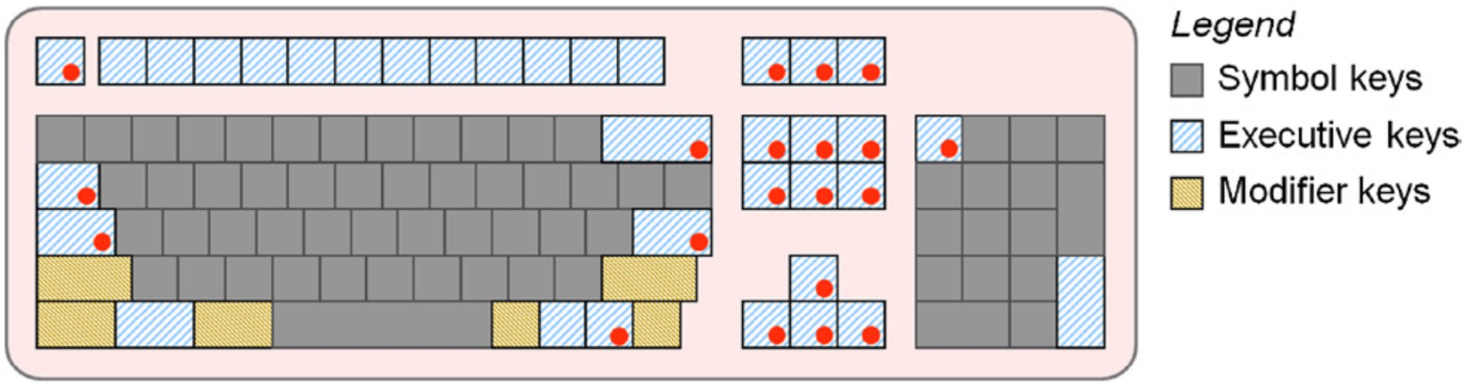


FIGURE 7.3

The key-action model (KAM) illustrated. Keys marked with (red) dots are executive keys that are not mirrored on both sides.

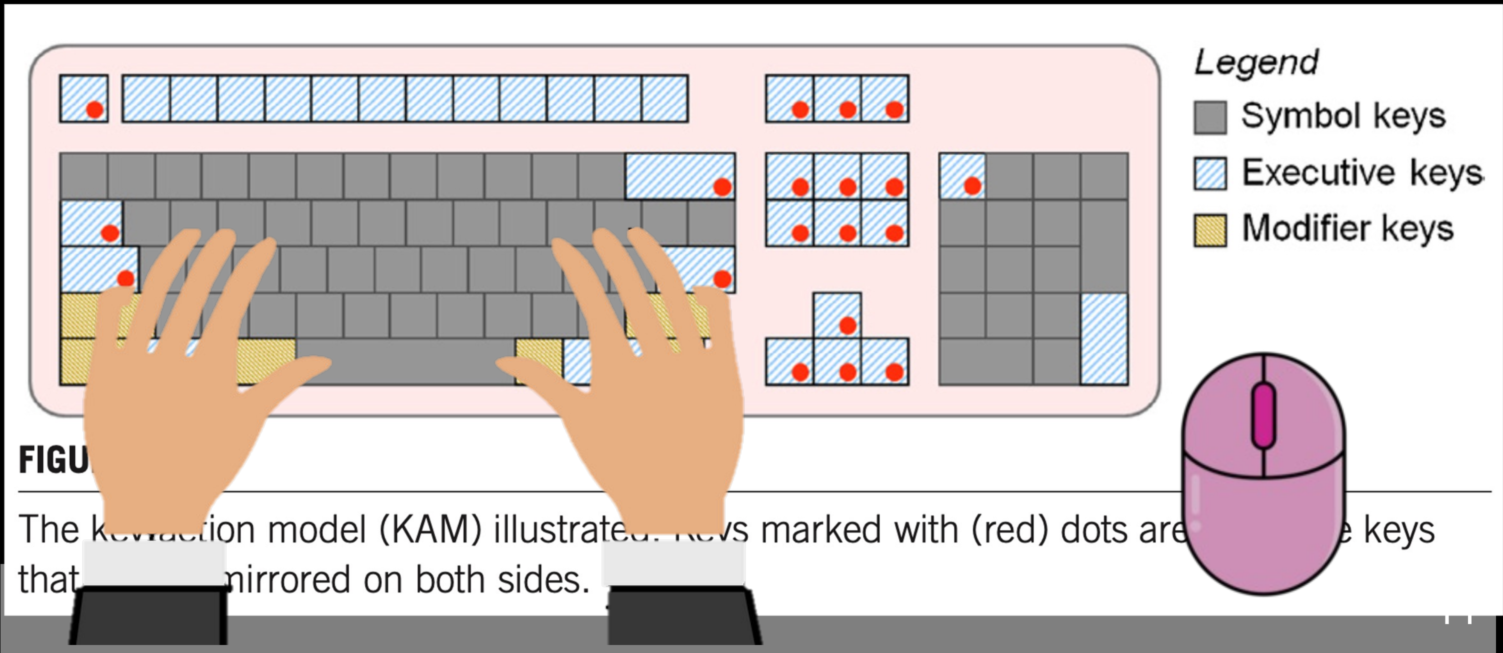
produce graphic symbols (e.g., A)

invoke actions (e.g., Esc)

modify effect of other keys (e.g., Ctrl)

Key-action model (KAM)

- What do you think of the KAM model?
- Can you think of additional categories to improve the model?
- Using this model, can you think of one problem in the desktop keyboard design?



FIGU

The k…ion model (KAM) illustrated. …s marked with (red) dots are … keys that … mirrored on both sides.

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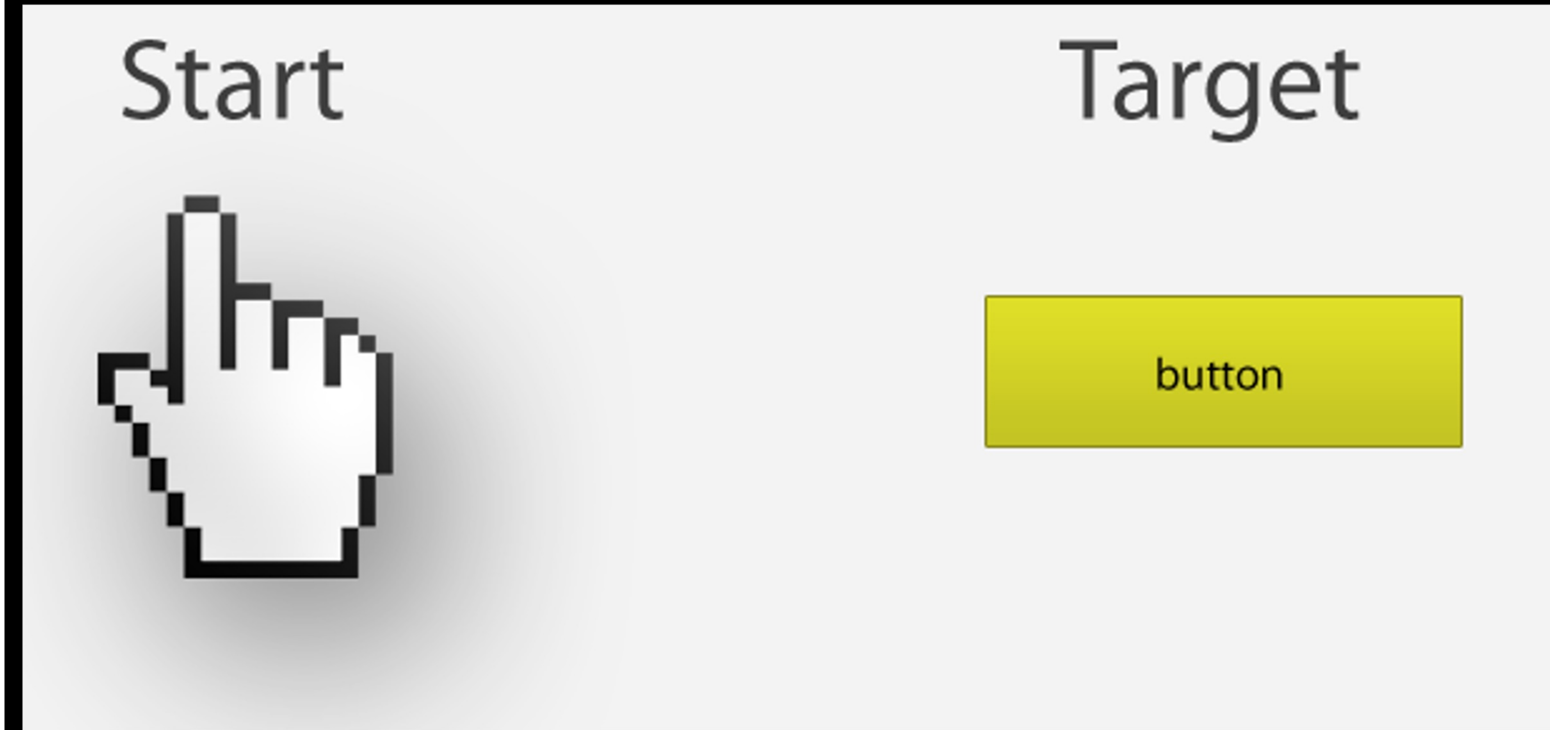
modify effect of other keys (e.g., Ctrl)

Predictive (Causal) models

- A predictive model is an equation.
- The equation predicts an outcome variable (or probability of the outcome) based on the value of one or more predictor variables.
- The equation can be simple (e.g., linear regression) or very complex.

- Useful to explain findings in past user experiments (model fitting).
- Useful to predict findings in future experiments (model validation).

Predictive modelling example: Target Selection



Examples of predictive models

- Fitts' Law:
  Predicts time to move to a target.

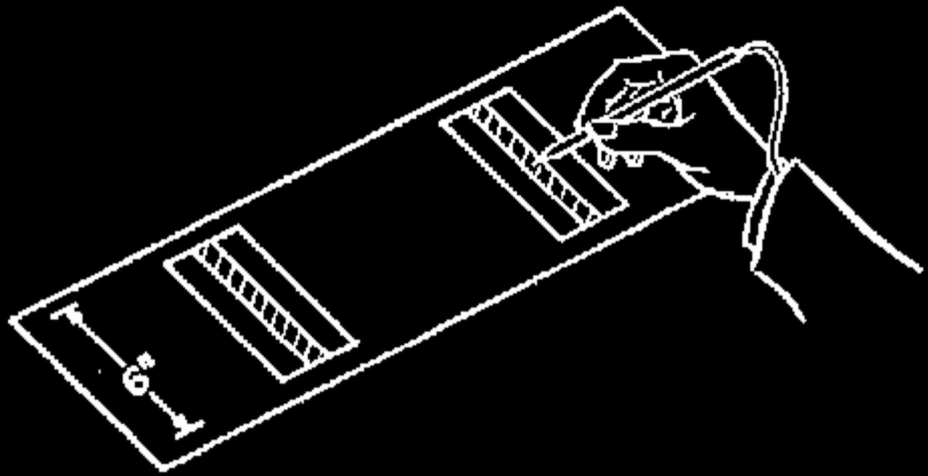
- Hick's Law:
  Predicts decision time based on number of choices.

- The Keystroke Level Model (KLM):
  Predicts time to complete tasks using low-level operations.

- Power law of practice:
  Predicts performance improvement with repetition.

Fitts' Law

- Developed by Paul Fitts in 1954
- Robust model of human psychomotor behavior
- Predicts movement time for rapid, aimed pointing tasks
- Fitts' discovery "was a major factor leading to the mouse's commercial introduction by Xerox“ [Stuart Card]



Fitts, Paul M. "The information capacity of the human motor system in controlling the amplitude of movement." *Journal of experimental psychology* 47.6 (1954): 381.

Card, Stuart K., English, William K., Burr, Betty J. (1978). *Ergonomics, 21(8):601–613* Evaluation of mouse, rate-controlled isometric joystick, step keys, and text keys for text selection on a CRT.

Fitts' Law

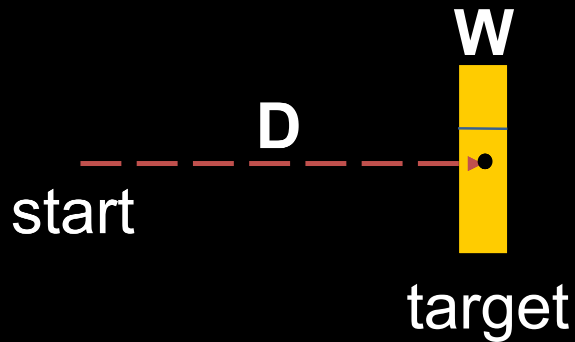
https://www.youtube.com/watch?v=R3TdamGweKw&ab_channel=NNgroup

Fitts' Law – Equation

- The time to acquire a target is a function of the distance to and size of the target and depends on the particular pointing system.

$$MT = a + b\boxed{\log_2\left(1+\frac{D}{W}\right)}$$

Index of Difficulty



- MT: movement time
- a and b: constants dependent on the pointing system
- D: distance to the target area
- W: width of the target

Fitts' Law – Equation

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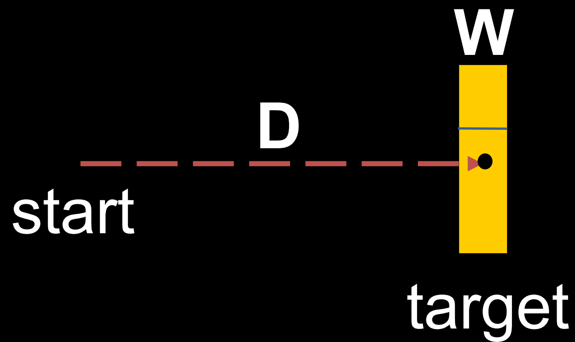
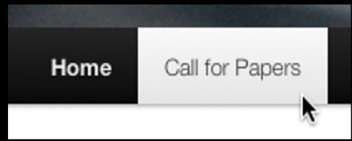


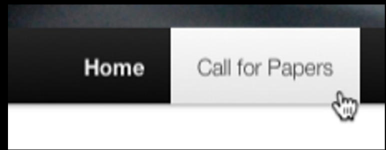
- MT: movement time
- a and b: constants dependent on the pointing system
- D: distance to the target area
- W: width of the target

Implications for HCI design

- Bigger buttons
  - e.g.. web links
  - e.g.. check / radio boxes
- Proportional to amount of use

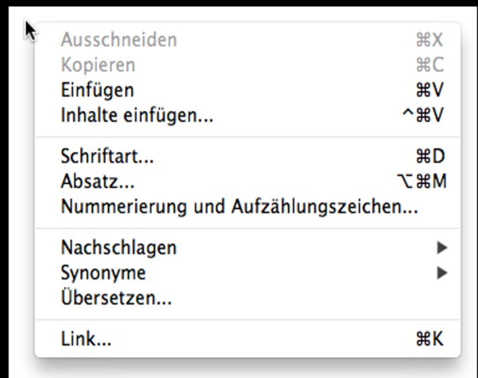


small target boundary



larger target boundary

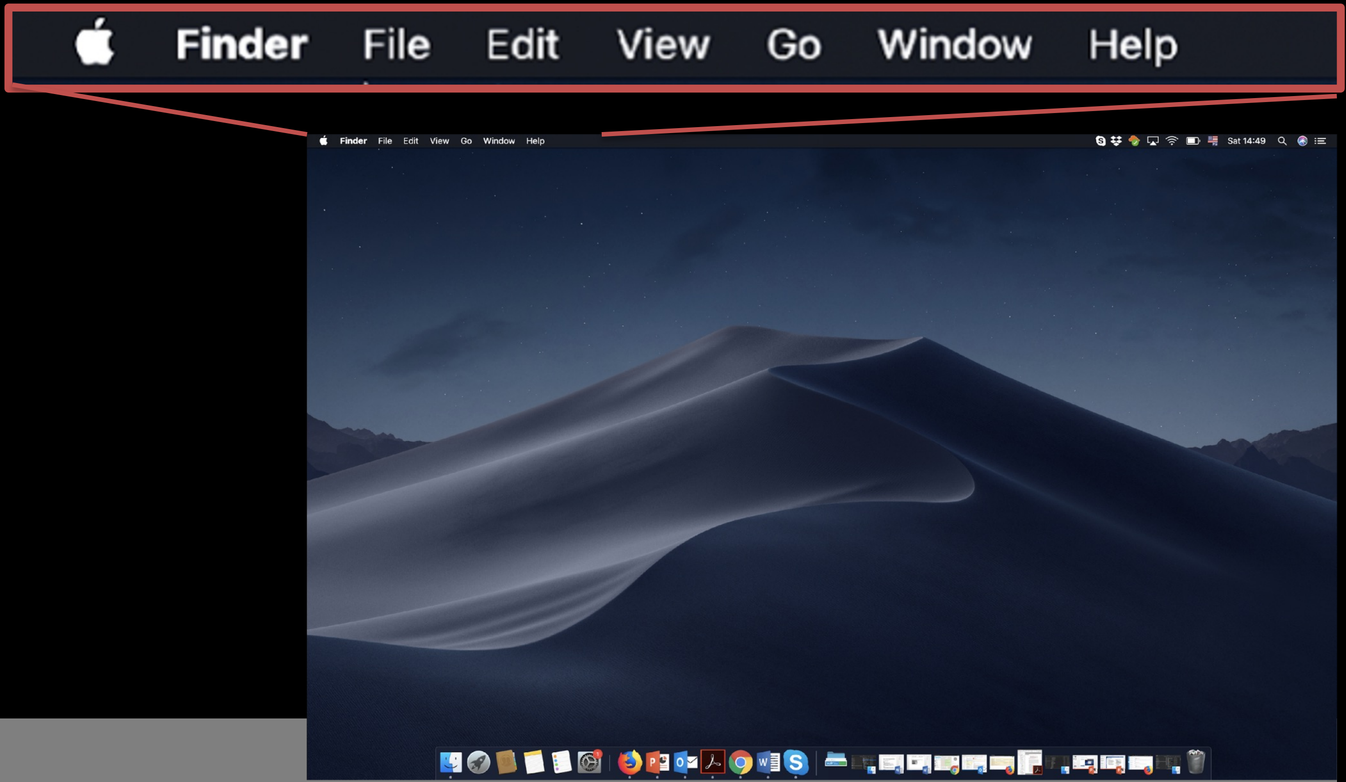
- Use current location of the cursor
  - distance is close to zero



Implications for HCI design

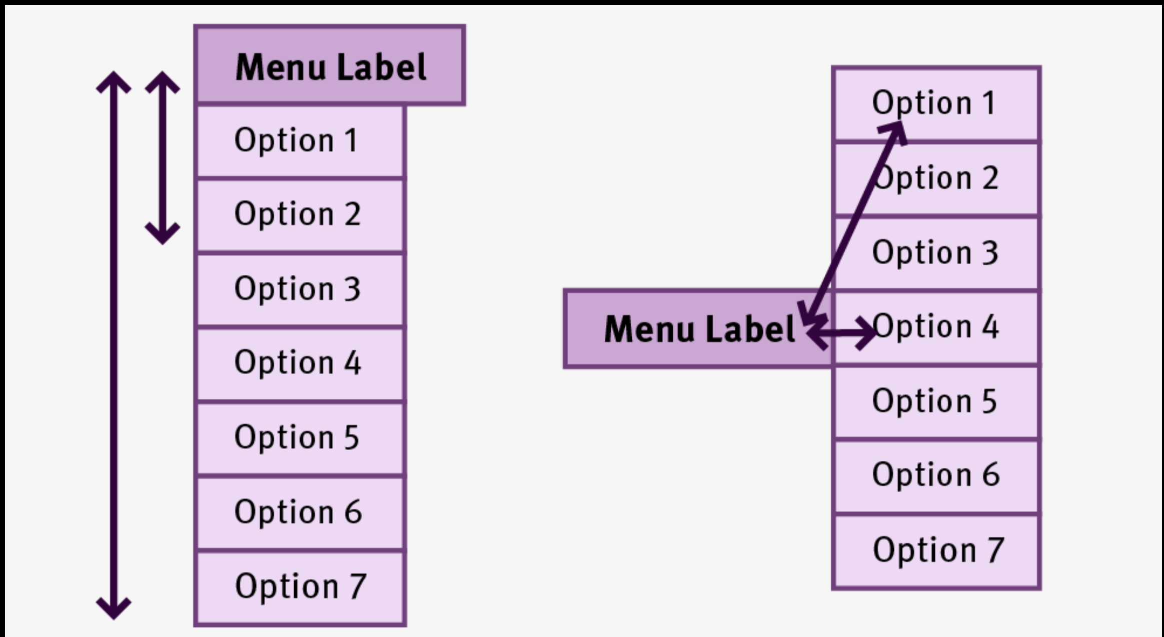
- Use edges and corners
  - corners have infinite height and width

macOS 10.14.3



Fitts' Law – Example question

- Which of the following menu designs is more efficient?



Hick's Law
(Hick-Hyman Law)

Let's do an experiment!

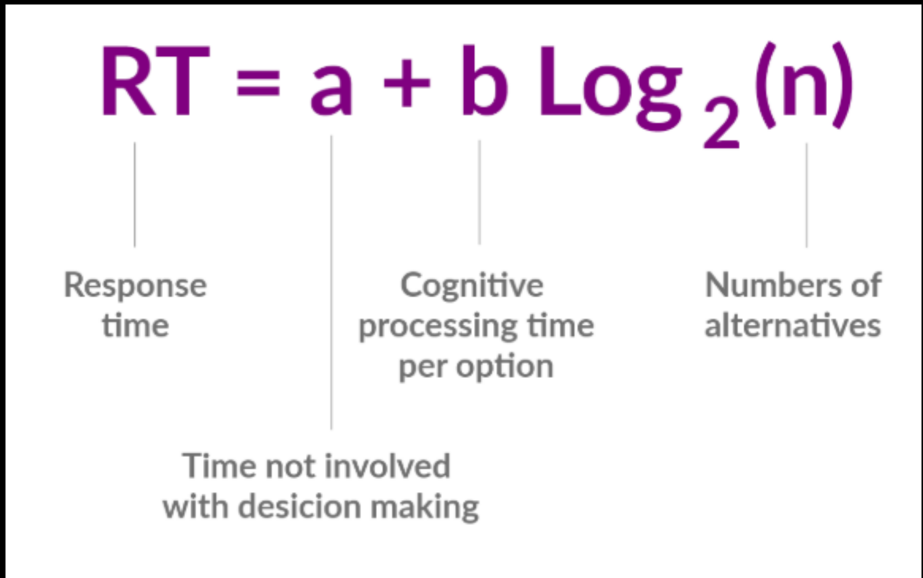
- You will see two lists.
- One is a list of cities and the other a list of country names.
- For each list, I'll ask you to find one item that I choose.
- Raise your hand as soon as you find it in the list.

- Tokyo
- Paris
- Detroit
- Jakarta
- Sydney
- Barcelona

- Luxembourg
- Malta
- Netherlands
- Poland
- Sweden
- United Kingdom
- Ireland
- Hungary
- Belgium
- Finland
- Czech Republic
- France
- Germany
- Greece
- Bulgaria
- Cyprus
- Slovakia
- Slovenia
- Latvia
- Portugal
- Austria
- Denmark
- Estonia
- Lithuania
- Spain
- Italy
- Romania

Hick's Law
(Hick-Hyman Law)

- The time needed to make a selection/decision is proportional to the log of the number of given choices.



- *a* ≈ 200 ms and *b* ≈ 150 ms/bit (Card et al., 1983)

Hick's Law (Hick-Hyman Law)

Let's change the experiment!

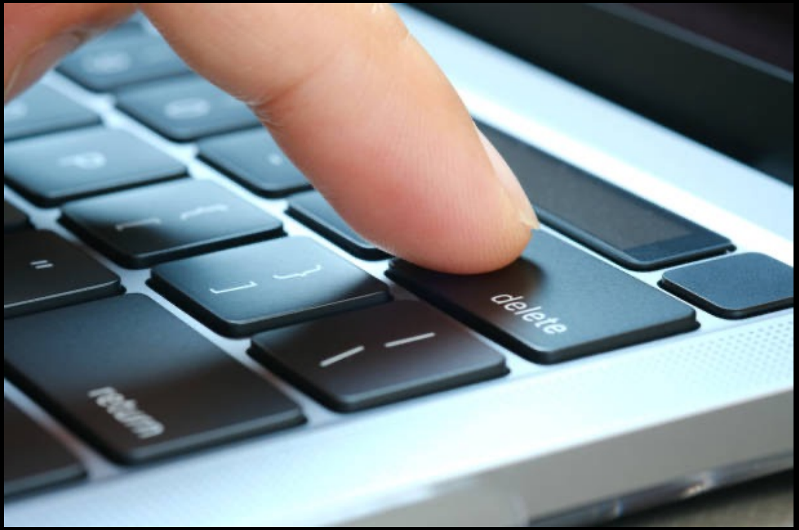
- Austria
- Belgium
- Bulgaria
- Cyprus
- Czech Republic
- Denmark
- Estonia
- Finland
- France
- Germany
- Greece
- Hungary
- Ireland
- Italy
- Latvia
- Lithuania
- Luxembourg
- Malta
- Netherlands
- Poland
- Portugal
- Romania
- Slovakia
- Slovenia
- Spain
- Sweden
- United Kingdom

- Luxembourg
- Malta
- Netherlands
- Poland
- Sweden
- United Kingdom
- Ireland
- Hungary
- Belgium
- Finland
- Czech Republic
- France
- Germany
- Greece
- Bulgaria
- Cyprus
- Slovakia
- Slovenia
- Latvia
- Portugal
- Austria
- Denmark
- Estonia
- Lithuania
- Spain
- Italy
- Romania

Hick’s Law
(Hick-Hyman Law)

https://www.youtube.com/watch?v=pbbTOzArcWQ&ab_channel=NNgroup

Keystroke Level Model (KLM)



- Developed by Card et al. in 1980s
- A simple method to measure task completion time
  - Unlike Fitts' and Hick's Laws that were borrowed from psychology, KLM was developed for HCI research.
- It assumes that each task can be divided into a sequence of low-level tasks or operators.
- Each operator is assigned a duration (amount of time a [*expert*] user would take to perform it).

Keystroke Level Model (KML)

List of KLM operators

K	Key press
B/BB	Mouse button press / click
P	Mouse pointing
D	Mouse drawing
H	Move hand between mouse and keyboard
M	Thinking (mental act)
R	Response time from the system

Keystroke Level Model (KML)

- Each operator has a determined execution time

Operator	Description	Associated Time
K	Keystroke, typing one letter, number, etc. or function key like 'CTRL', 'SHIFT'	Expert typist (90 wpm): 0.12 sec Average skilled typist (55 wpm): 0.20 sec Average non-secretarial typist (40 wpm): 0 .28 sec Worst typist (unfamiliar with keyboard): 1.2 sec
H	'Homing', moving the hand between mouse and keyboard	0.4 sec
B / BB	Pressing / clicking a mouse button	0.1 sec / 2*0.1 sec
P	Pointing with the mouse to a target	0.8 to 1.5 sec with an average of 1.1 sec Can also use Fitts' Law
$D(n_D, l_D)$	Drawing n_D straight line segments of length l_D	$0.9*n_D + 0.16*l_D$
M	Subsumed time for mental acts; sometimes used as 'look-at'	1.35 sec (1.2 sec according to [Olson and Olson 1995])
R(t) or ***W(t)***	System response (or 'work') time, time during which the user cannot act	Dependent on the system, to be determined on a system-by-system basis

Keystroke Level Model (KML) – Example 1

Delete a word in a text document using a keyboard

Let's say the user wants to delete the word "hello" in a text editor by selecting the word using the cursor and pressing Backspace five times. (hand is on the keyboard at the beginning)

Steps:

1. M – Think about the task
2. H – Move hand from keyboard to mouse
3. P – Point the cursor
4. B – tap the mouse (we assume double click)
5. H – Move hand back to keyboard
5. K × 5 – Press Backspace five times

Total time = M (1.35) + 2xH (2x0.4) + P (1.1) + B (0.1) + 5xK (5x0.28) = 4.75 seconds

Operator Time Values:

- Pointing ≈ 1.1 s
- Button = 0.1 s
- Homing = 0.4 s
- Mental = 1.35 s
- Key = 0.28 s
- Response = 1.0 s (system dependent)

Keystroke Level Model (KML) – Example 2

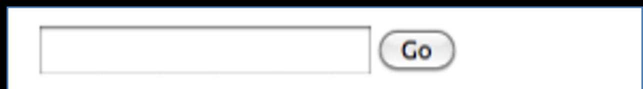
- A user wants to start the Photo application. Which of the following application launchers is the fastest?
  (Start: Hand rests on mouse)

Version 1:

Version 2:

Version 3:

Version 4:

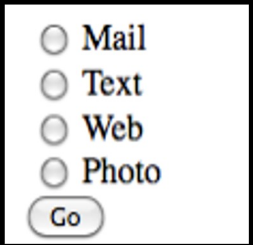


Operator Time Values:

- Pointing ≈ 1.1 s
- Button = 0.1 s
- Homing = 0.4 s
- Mental = 1.35 s
- Key = 0.28 s
- Response = 1.0 s (system dependent)

KLM – Example 2 (Results)

Version 1:



- M [Which app?]
- P [To Photo]
- B [Click]
- P [To Go]
- B [Click]
- R

$T = M + 2 \cdot P + 2 \cdot B + R = \mathbf{4{,}75\ s}$

Version 2:

- M [Which app?]
- P [To Photo]
- B [Click]
- R

$T = M + P + B + R = \mathbf{3{,}55\ s}$

Operator Time Values:

- Pointing ≈ 1.1 s
- Button = 0.1 s
- Homing = 0.4 s
- Mental = 1.35 s
- Key = 0.28 s
- Response = 1.0 s (system dependent)

KLM – Example 2 (Results)

Operator Time Values:

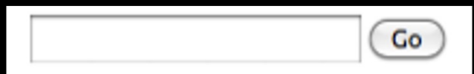
- Pointing ≈ 1.1 s
- Button = 0.1 s
- Homing = 0.4 s
- Mental = 1.35 s
- Key = 0.28 s
- Response = 1.0 s (system dependent)

Version 3:

- M [How to select?]
- P [To Drop-down]
- B [Click]
- M [Which app?]
- P [To Photo]
- B [Click]
- P [To Go]
- B [Click]
- R

$T = 2 \cdot M + 3 \cdot P + 3 \cdot B + R = \mathbf{7{,}30\ s}$

Version 4:



- M [App?]
- P [To Input]
- B [Click]
- H [To keyb.]
- 5·K [p-h-o-t-o]

Option 1:

H [To Mouse]
P [To Go]
B [Click]
R

Opt. 2:

K [↵]
R

$T_1 = M + 2 \cdot P + 2 \cdot B + 2 \cdot H + 5 \cdot K + R = \mathbf{6{,}95\ s}$

$T_2 = M + P + B + H + 6 \cdot K + R = \mathbf{5{,}63\ s}$

Summary
Descriptive vs. Predictive models

	Descriptive Models	Predictive Models
Purpose	Explain **what users do** and **how**	Predict **what will happen** and **how long**
Focus	Understanding, analysis	Measurement, forecasting
Type	Qualitative or procedural	Quantitative or mathematical
Use case	Designing and evaluating user interaction	Estimating performance (time, error, effort)
Output	Explanations, process maps	Numerical predictions
Example questions	"How do users complete this task?"	"How long will it take to click that button?"